Species traits influence the genetic consequences of river fragmentation on two co-occurring redhorse (*Moxostoma*) species

Scott M. Reid, Chris C. Wilson, Leon M. Carl, and Troy G. Zorn

Abstract: We used microsatellite DNA markers to test whether fragmentation of the Trent River (Ontario, Canada) has reduced genetic diversity and increased genetic differentiation among populations of river redhorse (*Moxostoma carinatum*) and shorthead redhorse (*Moxostoma macrolepidotum*). Allelic richness of both species was significantly greater along the free-flowing Muskegon River (Michigan, USA) than along the fragmented Trent River. Contrary to expectations, there was no evidence of a fragment length effect on genetic diversity, recent population bottlenecks, or increased relatedness among individuals in fragmented populations. High levels of linkage disequilibrium indicate extinction–recolonization population dynamics along the Trent River. For both species, pairwise F_{ST} tests identified weak but statistically significant population differentiation. In the Trent River, differentiation was significantly greater for river redhorse than for shorthead redhorse and, for both species, greater than in the Muskegon River. Moderate fragmentation effects likely reflect the permeability of the dam-lock system to redhorse movement. Differences between species indicate that as a result of smaller effective population sizes, habitat specialists and species at the periphery of their geographic range are more sensitive to river fragmentation.

Résumé : Des marqueurs microsatellites de l'ADN nous ont servi à tester si la fragmentation de la rivière Trent (Ontario, Canada) a réduit la diversité génétique et augmenté la différentiation génétique chez les populations de suceurs ballots (*Moxostoma carinatum*) et de suceurs rouges (*Moxostoma macrolepidotum*). La richesse allélique des deux espèces est significativement plus élevée dans le cours de la rivière Muskegon (Michigan, É.-U.) à écoulement libre que dans le cours fragmenté de la Trent. Contrairement à nos attentes, il n'y a aucune indication d'effet de la longueur d'un fragment sur la diversité génétique, ni de goulots d'étranglement récents dans la population, ni d'une parenté accrue entre les individus dans les populations fragmentées. De forts niveaux de déséquilibre de liaison indiquent l'existence d'une dynamique de population de type extinction–recolonisation le long de la Trent. Chez les deux espèces, les tests appariés F_{ST} identifient une différenciation des populations faible, mais statistiquement significative. Dans la Trent, la différenciation est significativement plus importante chez le suceur ballot que chez le suceur rouge et elle est plus forte, pour les deux espèces, que dans la Muskegon. Ces effets modérés de la fragmentation sont vraisemblablement le reflet de la perméabilité du système de barrages et d'écluses aux déplacements des suceurs rouges. Les différences entre les espèces indiquent que, comme résultat de leur taille de population réduite, les spécialistes de l'habitat et les espèces à la périphérie de leur aire géographique de répartition sont plus sensibles à la fragmentation des rivières.

[Traduit par la Rédaction]

Introduction

Hydrological and ecological changes associated with dams have contributed to the loss or reduction of migratory anadromous fish populations, smaller-bodied riverine fish species and invertebrates, and taxa dependent on flooding (Pringle et al. 2000). As a result of changes to population size and gene flow, dams can reduce genetic diversity within isolated fish populations and increase genetic differentiation among fish populations (Neraas and Spruell 2001; Yamamoto et al. 2004). In isolated populations, low genetic diversity can result from genetic drift, inbreeding, bottlenecks, and founder effects. At larger spatial scales, by restricting the movements of individuals, dams can increase genetic differentiation between previously connected or panmictic fish populations (Meldgaard et al. 2003), create upstream–downstream gradients in genetic diversity (Yamamoto et al. 2004), and disrupt the regional equilibrium between drift and immigration of new alleles in populations (Hutchison and Templeton 1999).

To date, research on the effects of riverine fragmentation on genetic diversity and population structure of fishes has been limited to single-species studies. Multispecies comparisons are needed to identify the common effect of landscape-level factors on genetic structure (Bermingham and Moritz

Received 10 April 2007. Accepted 8 February 2008. Published on the NRC Research Press Web site at cjfas.nrc.ca on 21 August 2008.
J19937

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 doi:10.1139/F08-093 © 2008 NRC Canada

1998; Castric and Bernatchez 2004) and to clarify the influence of species traits on the sensitivity of riverine fishes to landscape-level changes. The responses of aquatic and terrestrial organisms to habitat fragmentation are variable, often reflecting differences in species traits (Debinski and Holt 2000; Srikwan and Woodruff 2000; Cushman 2006). Generally, habitat specialists and (or) rare species are more sensitive to decreasing patch size and increasing isolation than habitat generalists and (or) common species (Ewers and Didham 2006). It is expected that genetic consequences of fragmentation will be influenced by ecological, demographic, and life-history characteristics that affect gene flow and subsequent population subdivision and genetic drift (Lesica and Allendorf 1995; Vucetich and Waite 2003; Zayed et al. 2005). For riverine fishes, among-species differences in population subdivision have been attributed to differences in spawning ecology, dispersal potential, and effective population size (Castric and Bernatchez 2004; Whiteley et al. 2004).

In this study, the genetic consequences of river fragmentation by dams on two redhorse species (river redhorse (*Moxostoma carinatum*) and shorthead redhorse (*Moxostoma macrolepidotum*)) were compared. Redhorse species (genus *Moxostoma*) are relatively large, laterally compressed, and superficially similar suckers (family Catostomidae) that are typically found in larger streams and rivers, but also in impoundments and lakes (Jenkins and Burkhead 1993; Scott and Crossman 1998). Redhorse species are good model species for investigating fragmentation effects as (*i*) they undertake seasonal migrations to spawning and overwintering habitats (Hackney et al. 1968; Grabowski and Isely 2006) and are therefore sensitive to barriers to movement, (*ii*) they have been reported to be sensitive to habitat changes related to dams (Quinn and Kwak 2003), (*iii*) growth, abundance, and reproductive characteristics of other sucker species have been correlated to habitat quantity, quality, and isolation (Trippel and Harvey 1987; Chen and Harvey 1999), (*iv*) species-level differences in regional distribution, abundance, life-history characteristics, and habitat preferences permit comparisons of sensitivity to habitat fragmentation, and (*v*) unlike sport fishes, the effects of dams will not be obscured by past stocking or high rates of fishing mortality.

The two redhorse species investigated in this study differ in their proximity to the core of their species range, habitat specialization, and population characteristics. Compared with shorthead redhorse, river redhorse has a small, discontinuous geographic range (Scott and Crossman 1998; NatureServe 2005), is near the northern edge of its geographic range in the Trent River (Scott and Crossman 1998), has a stronger association with free-flowing riverine habitats (Yoder and Beaumier 1986), and matures at later age (Reid 2006). In Canada, river redhorse is a species of special concern (Committee on the Status of Endangered Wildlife in Canada (COSEWIC) 2005) and is considered imperilled in neighbouring Great Lakes states (NatureServe 2005). These differences are expected to translate into different effective population sizes (Lesica and Allendorf 1995; Frankham 1996; Zayed et al. 2005) and, correspondingly, different sensitivities to fragmentation effects on gene flow.

Genetic diversity and population structure of river redhorse and shorthead redhorse in two Laurentian Great Lakes basin rivers with contrasting levels of fragmentation were characterized using microsatellite DNA. It was hypothesized that river fragmentation would increase population subdivision and reduce genetic diversity and that fragmentation effects would be more pronounced for river redhorse than for shorthead redhorse. The effect of fragmentation was evaluated by testing for evidence of (*i*) reduced genetic diversity and recent bottlenecks within fragmented populations, (*ii*) increased population subdivision, and (*iii*) an additive effect of dams on population differentiation. Demographic and mark–recapture data collected as part of a parallel study (Reid 2007) indicate that source–sink population dynamics (sensu Harrison 1991) operate along the Trent River. Spawning runs were absent from the smallest river fragments, and mark–recapture data indicate that a small percentage of individuals move between fragments, either upstream through the navigational locks or downstream over the dams (Reid 2007). Therefore, the persistence of small fragment populations is likely dependent on migrants from larger fragments. Individual assignment tests and a genetic-based framework (Tero et al. (2003) developed to evaluate models of movement along linearly arranged populations were applied to evaluate whether genetic data provide evidence of source–sink dynamics along the Trent River.

Materials and methods

Sample collection

Between 2001 and 2004, redhorse species were sampled from the Trent River, Ontario, and Muskegon River, Michigan, using a 5 kW pulsed DC boat-mounted electrofishing unit (Fig. 1). As part of the Trent–Severn Waterway, the Trent River has been highly fragmented by navigational and run-of-the-river hydroelectric dams. Ten river fragments were sampled along a 56 km section of the Trent River between the Hagues Reach Hydroelectric Generating Station and the Bay of Quinte, Lake Ontario. Dams and locks along this reach of the Trent River were constructed during 1838 to 1839 and 1911 to 1913 (Angus 1998). Both species have not been captured upstream of the study reach. For comparison, eight sites were sampled along a 76 km free-flowing reach of the Muskegon River between the Croton Hydroelectric Dam (constructed in 1905) and Muskegon Lake. In contrast to the Trent River, the Muskegon River, a Lake Michigan tributary, has few natural or constructed barriers along its mainstem.

The likelihood of redhorse capture was maximized by targeting shallow, fast-flowing habitats with coarse bed material (gravel, cobble, and boulder) and run or pool habitats immediately downstream (Yoder and Beaumier 1986). Along the Trent River, these habitats were located in tailwater areas downstream of dams. Sampling site selection along the Muskegon River was based on the location of large riffle areas and on boat access. Fin clips were taken and stored either in 95% ethanol or as dried tissue in scale sample envelopes. Redhorses sampled were adult-sized (shorthead redhorse, 40–62 cm total length; river redhorse, 50–80 cm total length) and represented multiple age classes.

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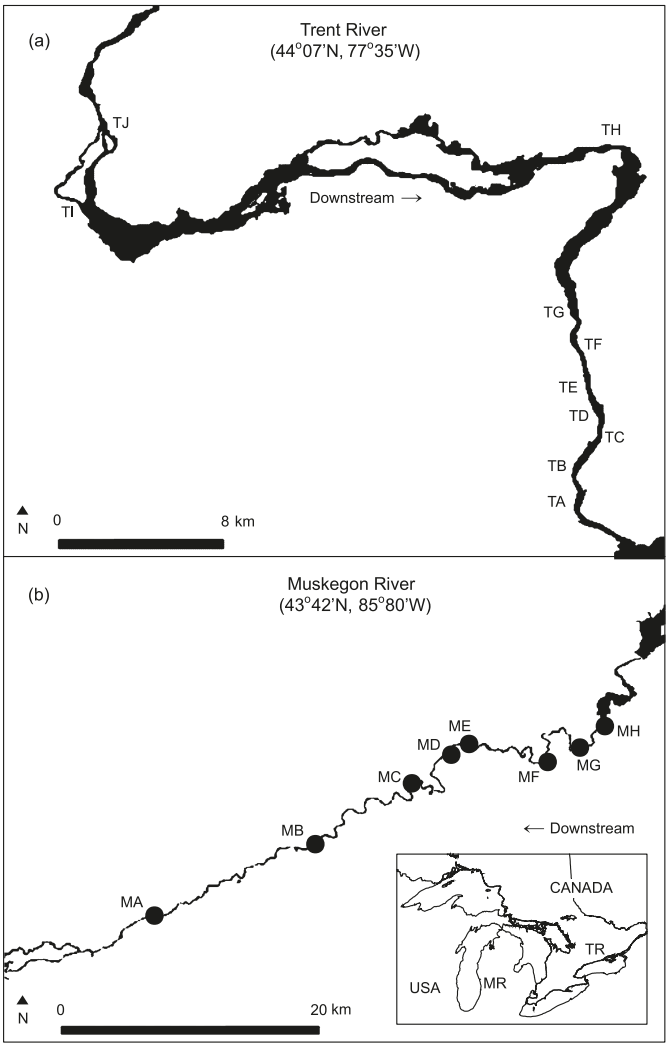
Fig. 1. Locations of redhorse sampling sites along (*a*) the Trent River (Ontario) and (*b*) the Muskegon River (Michigan). River locations within the Laurentian Great Lakes basin are identified in the inset map (TR, Trent River; MR, Muskegon River).



Laboratory methods

DNA was extracted from air-dried and 95% ethanol-preserved fin clips. Tissue samples were placed in a 96-deep-well plate and lysed with 400 μL of lysis buffer (50 mmol·L^{-1} Tris ph 8, 100 mmol·L^{-1} NaCl, 1 mmol·L^{-1} EDTA, 1% SDS *w*/*v*) with 1.5 μL Proteinase K (20 mg·mL^{-1}). To precipitate the DNA, 16 μL of 5 mol·L^{-1} NaCl and 800 μL of 80% isopropanol were added to each well. Samples were then centrifuged for 45 min at 6200 rpm to pellet the DNA. Supernatant was removed and the pellet was washed with 1000 μL of 70% ethanol to remove excess salts. Plates were vortexed and recentrifuged for 45 min at 6200 rpm; ethanol was carefully removed and plates were left to dry until all ethanol had evaporated. DNA was resuspended in 100–200 μL of 1× TE (10 mmol·L^{-1} Tris, 1 mmol·L^{-1}EDTA). DNA yield and quality was visually assessed by horizontal electrophoreses on a 1.5% agarose gel alongside molecular mass ladders.

Eleven microsatellite DNA loci were amplified using the polymerase chain reaction (PCR) (primer sequences, allele size ranges provided in Supplemental Table S1.[2] Microsatellite primers used were initially developed for Lost River sucker (*Deltistes luxatus*) (Tranah et al. 2001), copper redhorse (*Moxostoma hubbsi*) (Lippe et al. 2004), and robust redhorse (*Moxostoma robustum*) (I. Wirgin, New York University Medical Center, Nelson Institute of Environmental Medicine, Tuxedo, NY 10987, USA, unpublished primers). Over 120 potential polymorphic dinucleotide and tetranucleotide primers have been identified for these three catostomid species. Based on the reported levels of heterozygosity and allele number, a subset of 28 microsatellite primers was evaluated. Suitability was based on the level of polymorphism among individuals and whether primers would reliably amplify. For each primer, four to five samples of each species were amplified. Test samples for both species were collected from the Grand River (Lake Erie basin), Muskegon River, and Trent River. The potential for cross reactivity (i.e., primer–dimer formation) of individual primer pairs was evaluated using AutoDimer (Vallone and Butler 2004). In contrast to Lippe et al. (2004), *Mohu-lav*203, *Mohu-lav*212, and *Mohu-lav*213 were not polymorphic for river redhorse and shorthead redhorse. Lost River sucker (*dlu*) and robust redhorse (RR) loci that were monomorphic, or poorly amplified, included *dlu*434, *dlu*467, *dlu*4139, *dlu*4235, *dlu*4276, *dlu*4300, *dlu*4314, *dlu*4315, RR15, RR24, RR33, RR37, and RR44.

Loci were co-amplified in duplex and triplex reactions (simultaneous amplification of two or three loci in one reaction). For their subsequent detection, each microsatellite locus was amplified using a fluorescent-dye-labelled primer along with an unlabelled primer (Table 1). Polymerase chain reactions were performed in 11 μL total volumes containing 1.5 μL of DNA template, 1.1 μL PCR buffer, 0.8 μL of $MgCl_2$ (25 mmol·L^{-1}), 0.24 μL of dNTPs (10 μmol·L^{-1}), 0.13 μL of *Taq* I DNA polymerase (5U·μL^{-1}], microsatellite primers (10 μmol·L^{-1}) (volumes listed below), and 5.6–6.0 μL of ddH_2O (depending on primer volumes within each multiplex). In the duplex reaction (MULTI_1), primer volumes were 0.05 μL RR13 (FAM) and 0.1 μL RR55 (NED). In the first triplex (MULTI_2), primer volumes were 0.16 μL RR38 (HEX), 0.13 μL *dlu*4184 (NED), and 0.16 μL *dlu*4296 (FAM). Primer volumes in the second triplex (MULTI_3) were 0.1 μL *Mohu-lav* 198 (NED), 0.1 μL *Mohu-lav* 296 (FAM), and 0.2 μL *Mohu-lav* 305 (HEX). Primer volumes in the third triplex (MULTI_4) were 0.10 μL *dlu*45 (FAM), 0.21 μL *dlu*405 (HEX), and 0.10 μL *Mohu-lav*194 (NED). Reactions were amplified in a thermocycler (model PTC-100; MJ Research, Inc., Waltham, Mass.) using an initial denaturation at 94 °C (11 min), followed by 35 cycles of 94 °C (1 min), 55 °C (MULTI_1 and MULTI_2) or 60 °C (MULTI_3 and MULTI_4) (1 min), and 72 °C (1 min), with a final extension at 60 °C (45 min).

[2] Supplementary data for this article are available on the journal Web site (http://cjfas.nrc.ca) or may be purchased from the Depository of Unpublished Data, Document Delivery, CISTI, National Research Council Canada, Building M-55, 1200 Montreal Road, Ottawa, ON K1A 0R6, Canada. DUD 3794. For more information on obtaining material, refer to http://cisti-icist.nrc-cnrc.gc.ca/cms/unpub_e.html., available online from NRC Data Depository).

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Table 1. Site details and associated measures of abundance and genetic diversity for river redhorse (*Moxostoma carinatum*) and shorthead redhorse (*Moxostoma macrolepidotum*) sampled from the Trent River, Ontario.

			M. carinatum				*M. macrolepidotum*			
Site	rkm[a]	Fragment length (km)	n	Maximum spring CPUE[b]	A_R	H_O	n	Maximum spring CPUE[b]	A_R	H_O
TA	3	3.1	41	39.3	4.40	0.74	48	31.7	8.73	0.78
TB	4.1	1.1	9	1.7	4.36	0.77	43	7.7	8.67	0.78
TC	6.6	2.5	47	15.0	4.25	0.71	51	27.3	8.78	0.74
TD	7.5	0.9	12	4.5	4.02	0.73	21	9.6	8.26	0.73
TE	8.8	1.3	4	1.3	4.09	0.61	11	1.7	8.52	0.69
TF	11	2.2	47	5.7	4.32	0.70	54	18.3	8.59	0.74
TG	13.4	2.4	15	3.0	4.21	0.79	57	9.7	8.86	0.77
TH	23.6	10.2	48	41.3	4.21	0.71	53	37.0	8.30	0.79
TI	46.6	23.0	56	11.3	4.17	0.71	55	23.7	8.91	0.80
TJ	48.9	2.3	35	4.0	4.00	0.71	30	5.7	7.86	0.68
All			314		4.40	0.72	423		8.97	0.76

Note: See Fig. 1 for site locations. CPUE, catch per unit effort; A_R, standardized allelic richness; H_O, observed heterozygosity.

[a]River kilometres (rkm) is the distance in kilometres upstream from river mouth.

[b]Number of individuals per 1000 seconds of boat electrofishing.

After amplification, PCR products were diluted in 11 μL of sterile deionized water and then combined with an internal lane standard composed of formamide, loading buffer, and ROX 350 size standard (Applied Biosystems Inc., Foster City, Calif.). Diluted PCR products were denatured at 96 °C for 2 min and then loaded into a 5% Long Ranger acrylamide gel and electrophoresed on an ABIPrism 377 DNA sequencer. Reactions were visualized using GeneScan software (Applied Biosystems Inc.), and bands were scored by comparison with standard ROX 350 ladder using GenoTyper (Applied Biosystems Inc.) and visual proofreading. Genotypes were checked for scoring errors that might be attributable to stutter-products, large allele dropout, or the presence of null alleles using MICROCHECKER (version 2.2; van Oosterhout et al. 2004).

Data analysis

Patterns of allelic diversity and disequilibrium

The number of alleles (N_A), standardized allelic richness (A_R), observed heterozygosity (H_O) statistics, and allele frequencies were compiled using FSTAT (version 2.9.3; Goudet 2001) and the Microsoft Excel Microsatellite Toolkit (Park 2001). Differences in A_R among sites and among loci were each tested using one-way analysis of variance (ANOVA). Conformation of A_R data to the assumption of normality was assessed with the Shapiro–Wilk test. Analysis of molecular variance (AMOVA) was completed using GenAlEx 6 (Peakall and Smouse 2006). Genotypes at each locus for each population were tested for conformance to Hardy–Weinberg equilibrium (HWE) using GENEPOP (version 3.4; Raymond and Rousset 1995). Fisher's exact test of linkage disequilibrium was performed between all pairs of loci with 10 000 dememorizations, 10 000 batches, and 10 000 iterations per batch. To address the risk of inflated type I errors resulting from multiple tests, significance levels ($p < 0.05$) were adjusted using the Bonferroni correction (Rice 1989).

The paired-sample t test was used to test the hypothesis that genetic diversity (A_R and H_O) of the more abundant shorthead redhorse was greater than that of the river redhorse (Zayed et al. 2005). For river redhorse, *dlu*405 (both rivers) and *Mohu-lav*198 (Muskegon River only) did not amplify consistently and (or) were monomorphic. Therefore, the number of loci available for analysis was not consistent across species and rivers. All tests comparing genetic diversity and population differentiation between species and rivers were standardized to the same number of microsatellite loci using FSTAT.

In the Trent River, river fragment length and redhorse abundance are positively correlated (Reid 2007). Populations in short river fragments were therefore hypothesized to have comparatively low levels of genetic diversity. To test for a river fragment size effect on genetic diversity, Spearman's rank correlations (r_S) were calculated between genetic diversity measures (A_R and H_O) and (*i*) river fragment length and (*ii*) maximum spring catch-per-unit-effort (CPUE, an indicator of the number of spawners). By facilitating downstream dispersal and limiting upstream dispersal, the unidirectional downstream flow of water has been hypothesized to create upstream–downstream gradients of genetic diversity (Hernandez-Martich and Smith 1997). Correlations between genetic diversity measures and distance upstream from river mouth were therefore also calculated for both rivers.

The relationship between river fragment length and genetic diversity (A_R and H_O) was also tested by comparing populations grouped into three length classes: small (0.9–1.3 km; n = 3); medium (2.2–2.4 km; n = 4); and large (>3.1 km; n = 3). Fragment length classes differ in the amount of available shallow tailwater habitat (small, ~50–100 m; medium, ~500–700 m; large, >1.0 km) and the demographic characteristics of populations (e.g., number and sex ratio of spring spawners) (Reid 2007). Differences between river fragment length classes were tested with randomization tests (1000 permutations) as implemented in FSTAT. Randomization tests were also used to compare the genetic diversity (A_R and H_O) of redhorse populations in the fragmented Trent River to Muskegon River populations.

By creating small populations that are isolated from im-

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migration and preferred spawning habitats, river fragmentation could trigger population bottlenecks and result in low genetic diversity through either genetic drift or inbreeding (Frankham et al. 2002). Redhorse populations within Trent River fragments were tested for evidence of recent population bottlenecks and inbreeding. Recently bottlenecked populations may exhibit gametic disequilibrium (Waples 2002), reduced genetic diversity (particularly the loss of rare alleles) (Allendorf 1986), and increased heterozygosity relative to that expected at mutation–drift equilibrium (Cornuet and Luikart 1996). Evidence of recent and pronounced reductions in population size was tested using BOTTLENECK (version 1.2.02; Piry et al. 1999) and by calculating Garza and Williamson's (2001) M statistic. When a population undergoes a reduction in its effective population size, allelic diversity is reduced more rapidly than heterozygosity; as a consequence, there is a transient deficiency in the number of alleles (Cornuet and Luikart 1996). BOTTLENECK compares a population's heterozygote excess to that expected to be found at mutation–drift equilibrium. As microsatellite evolution is more similar to the stepwise mutation model (SMM) than the infinite allele model (IAM), a two-phase mutation model with 95% SMM, 5% IAM, and 12% variance of multiple-step mutations was assumed. Simulation iterations (5000) were conducted for each population (Cornuet and Luikart 1996; Piry et al. 1999). Simulations were limited to sites with at least 20 individuals sampled (Luikart and Cornuet 1998). Significance of heterozygote excess over all loci was tested with the Wilcoxon's signed-rank test. Garza and Williamson's (2001) M is the mean of the ratio of the total number of alleles (k) and the number of repeats over the allele size range (r) for each locus. The ratio is based on the expectation that when the effective size of a population is reduced and drift occurs, reductions in allele size range will occur more slowly than the loss of individual alleles. Based on simulations and data from natural populations, values of $M < 0.68$ are indicative of populations that have undergone a recent bottleneck (Garza and Williamson 2001).

To determine globally whether individual redhorse collected from each site were more related than expected under random mating, a Monte Carlo resampling test (IDENTIX; Belkhir et al. 2002) that uses a multilocus estimator ($r_{x,y}$) for pairwise comparisons (Queller and Goodnight 1989) was applied. High relatedness values and (or) excess representation by a few families could be indicative of inbred individuals in a population (Lippe et al. 2006). Tests of both the mean and variance of relatedness coefficients were undertaken, as significantly higher variance can indicate that several independent groups of related individuals were sampled. As several populations did not conform to Hardy–Weinberg expectations, resampling (1000 permutations) was carried out at the genotype level.

Population structure

Differentiation among populations was evaluated using both population-level (F_{ST}) and individual assignment methods. Global F_{ST} and pairwise F_{ST} were estimated using Weir and Cockerham's θ (Weir and Cockerham 1984). F_{ST} values and significance of estimates were calculated with FSTAT. F_{ST} was used instead of R_{ST}, as F_{ST} estimates are more conservative when relatively few (<20) microsatellite loci are used and when populations have recently diverged (Gaggiotti et al. 1999). Tests for population differentiation were limited to sites with at least 20 genotyped individuals.

Genetic distances among population pairs were estimated by computing Cavalli-Sforza and Edwards (1967) genetic chord distances (D_{CE}) in POPULATIONS (version 1.2.28; Oliver Langella, unpublished (1999), http://bioinformatics.org/~tryphon/populations/). D_{CE} was used because it is commonly applied in studies of fish population structure, does not assume any mutation model, and performs well in simulations with microsatellite data (Takezaki and Nei 1996). To visualize the genetic relationships among sites, D_{CE} chord distances were used to build an unrooted neighbour-joining tree in TREEVIEW (version 1.66; Page 1996). Robustness of the typology was evaluated by 1000 bootstrap resamplings as implemented in POPULATIONS.

The effect of dams on genetic differentiation was evaluated by testing for significant correlations between matrices of pairwise genetic distance (F_{ST} and D_{CE} values), geographic distance, and number of dams separating population pairs. As geographic distances between population pairs were greater than habitat width, F_{ST} values were adjusted with the following transformation: $F_{ST}/(1 - F_{ST})$ (Rousset 1997). Significance of correlations was based on Mantel and partial Mantel tests (10 000 permutations) calculated using the software IBD (Bohonak 2002). Using pairwise F_{ST} and D_{CE} estimates from the same set of populations, the paired-sample t test was used to test the hypothesis that genetic differentiation between river redhorse population pairs was greater than for shorthead redhorse populations.

Two approaches were applied to evaluate whether fragmented Trent River populations were in regional equilibrium (i.e., a balance between the loss of alleles due to drift and their replacement by gene flow) or whether genetic drift was more influential than gene flow. Assuming a stepping-stone model of regional population structure, Hutchison and Templeton (1999) interpreted populations of a region to be at equilibrium (CASE I) if (*i*) there is a significant relationship between pairwise F_{ST} genetic and geographic distances separating populations and (*ii*) a scatterplot of pairwise F_{ST} and geographic distances reveals a positive and monotonic relationship over all distance values of a region. The second approach, using a Markov Chain Monte Carlo (MCMC) approach implemented in 2mod (Ciofi et al. 1999; software available from http://www.rubic.rdg.ac.uk/~mab/software.html), calculates the relative likelihood of two alternate models of demographic history: a model of immigration–drift equilibrium versus a model of drift (without equilibrium). The MCMC search was carried out twice for 10^5 iterations with the initial 30 000 discarded as burn-in.

Tero et al. (2003) presented a genetic-based framework to evaluate five hypothetical models of migration along linearly arranged populations. The five models include (*i*) a genetically uniform population model with free gene flow between populations, (*ii*) a fragmented population model without gene flow, (*iii*) a stepping-stone population model, (*iv*) a source-sink model, and (*v*) the classic metapopulation model. The applicability of the different models to given sets of populations is evaluated based on gene flow, population differences in genetic diversity, mutation–drift equili-

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brium, and extinction–replacement dynamics. Characteristics that would support a source–sink population model for the Trent River include gene flow from source to sink populations, greater genetic diversity in source populations relative to sink populations, and high levels of linkage disequilibrium within populations and for the total population set (Tero et al. 2003).

Individual assignment tests of multilocus genotypes were conducted to assign probable source populations as origins of individual redhorse sampled from hypothesized sink populations. Assignment tests were also done to evaluate membership within source populations. Individuals were assigned to a population based on log-likelihood probabilities of their genotypes occurring by chance in potential source populations using GeneClass (version 2; Cornuet et al. 1999; Piry et al. 2004). The Bayesian leave-out method (i.e., remove an individual's genotype from the data set before calculating its log-likelihood probability of assignment to population group) was used for self-assignments of individuals to populations (Rannala and Mountain 1997). Confidence in individual assignments was based on 10 000 simulations using Monte Carlo resampling methods developed by Paetkau et al. (2004). The rejection level was set at $p < 0.01$. Because of heterozygote deficiency across most populations, *dlu*4296 (shorthead redhorse) and *dlu*45 (river redhorse) were not included in simulations.

Results

Genetic diversity and disequilibrium

The microsatellite loci utilized in this study were moderately to highly polymorphic for both river redhorse and shorthead redhorse (Tables 1 and 2). Allele size frequencies for each microsatellite locus and population are presented in Supplemental Appendices S1–S4.[2] Tests of heterozygote deficiency showed deviations from HWE in 12%–25% of the population–locus combinations. Deviations were not consistent across loci within populations, but instead were associated with individual loci across most populations sampled. The following loci were consistently out of HWE: *dlu*45 (river redhorse: six Trent River and six Muskegon River populations); *dlu*4184 (shorthead redhorse: seven Muskegon River populations); *dlu*4296 (shorthead redhorse: 10 Trent River and six Muskegon River populations); and RR55 (river redhorse: five Muskegon River populations).

Linkage disequilibrium among pairs of loci tested within each population was present in less than 2.2% of combinations. No loci pair or within-population trends were evident. After pooling populations, linkage disequilibrium occurred for a greater percentage of combinations, mostly associated with the Trent River. For Trent River samples, linkage disequilibrium occurred at 22% of river redhorse and 24% of shorthead redhorse loci combinations. For Muskegon River samples, linkage disequilibrium occurred at 5% of river redhorse and 7% of shorthead redhorse loci combinations.

Allelic richness of Muskegon River redhorse was significantly greater than that of redhorse sampled from the fragmented Trent River (randomization test: river redhorse $p = 0.027$; shorthead redhorse $p = 0.001$). In contrast, observed heterozygosity was not significantly different between rivers (randomization test: river redhorse $p = 0.21$; shorthead redhorse $p = 0.52$). Genetic diversity of shorthead redhorse populations in the Trent River was significantly greater than that of river redhorse (one-tailed paired t test: A_R, $t = 3.56$, df = 9, $p = 0.003$; H_O, $t = 2.47$, df = 9, $p = 0.018$). For the Muskegon River, only shorthead redhorse allelic richness was significantly greater than river redhorse (one-tailed paired t test: A_R, $t = 12.4$, df = 6, $p < 0.0001$; H_O, $t = -0.68$, df = 6, $p = 0.26$).

Table 2. Site details and associated measures of genetic diversity for river redhorse (*Moxostoma carinatum*) and shorthead redhorse (*Moxostoma macrolepidotum*) sampled from the Muskegon River, Michigan.

		M. carinatum			*M. macrolepidotum*		
Site	rkm[a]	n	A_R	H_O	n	A_R	H_O
MA	22.8	12	6.54	0.79	47	12.90	0.76
MB	37.5	7	7.01	0.77	18	13.55	0.82
MC	53.8	28	6.84	0.75	48	13.04	0.74
MD	62.2	—	—	—	50	12.89	0.78
ME	66.1	34	6.93	0.77	48	13.68	0.76
MF	77.4	47	6.98	0.79	45	13.09	0.73
MG	87.7	30	6.89	0.75	43	13.09	0.80
MH	90.3	38	7.15	0.80	48	13.06	0.75
All		196	7.30	0.77	347	13.55	0.76

Note: See Fig. 1 for site locations. A_R, standardized allelic richness; H_O, observed heterozygosity.

[a]River kilometres (rkm) is the distance in kilometres upstream from river mouth.

The amount of genetic variability attributed to among-population differences was greater in the fragmented Trent River than in the Muskegon River (Table 3). However, AMOVA indicated that most genetic variability occurred within populations rather than among populations. For both species and rivers, allelic richness was significantly different among loci (Shapiro–Wilk: $p > 0.05$; ANOVA: $F = 43.1$ to 74.5, $p < 0.001$), but not among sites (Shapiro–Wilk: $p > 0.05$; ANOVA: $F = 0.08$ to 0.25, $p > 0.05$). Additionally, no significant differences in A_R and H_O were detected between different fragment length classes (randomization test: p values > 0.16). Shorthead redhorse heterozygosity in the Trent River was positively correlated with maximum spring CPUE ($r_S = 0.68$, $p = 0.047$) but not with river fragment length or distance upstream from river mouth. Other correlations between genetic diversity measures and fragment length, maximum spring CPUE, and distance upstream from river mouth were not significant ($p > 0.05$).

Evidence of recent population bottlenecks was not detected within fragmented Trent River populations. Wilcoxon signed-rank tests for heterozygote excess were not significant (p values = 0.79–0.99), indicating no significant departure from mutation–drift equilibrium. For both species, M was greater (river redhorse, 0.70–0.80; shorthead redhorse, 0.74–0.92) than the threshold value ($M < 0.68$) proposed by Garza and Williamson (2001). Values for Trent River populations were equal to or greater than those calculated for the unfragmented Muskegon River (river redhorse, 0.69–0.73; shorthead redhorse, 0.74–0.87). For Trent River redhorse populations, simulations based on the mean (p values > 0.09) and variance (p values > 0.25) of relatedness estimates

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Table 3. Analysis of molecular variance (AMOVA) results and global F_{ST} values for river redhorse (*Moxostoma carinatum*) and shorthead redhorse (*Moxostoma macrolepidotum*) populations in the Trent and Muskegon rivers.

River	Species	Individual (%)	Population (%)	Φ	p value	Global F_{ST}
Trent	*M. carinatum*	94	6	0.059	0.0001	0.034
	M. macrolepidotum	98	2	0.025	0.001	0.014
Muskegon	*M. carinatum*	99	1	0.012	0.002	0.007
	M. macrolepidotum	99	1	0.008	0.001	0.009

Note: Percent distribution of allelic diversity among individuals within populations and among populations is presented.

Table 4. Summary of pairwise tests of genetic differentiation among Trent River river redhorse (*Moxostoma carinatum*; below diagonal) and shorthead redhorse (*Moxostoma macrolepidotum*; above diagonal) populations.

	TA	TB	TC	TF	TG	TH	TI	TJ
TA		0.009	0.009	0.013	0.008	0.014	0.005	0.026
TB	—		0.024	0.027	0.022	0.034	0.015	0.041
TC	**–0.002**			0.005	**0.003**	0.010	0.010	0.020
TF	0.019	—	0.025		**0.001**	0.006	**0.005**	0.019
TG	—	—	—	—		0.008	0.007	0.014
TH	0.029	—	0.034	**0.008**	—		0.012	**0.006**
TI	0.035	—	0.035	0.035	—	0.064		0.023
TJ	0.046	—	0.051	0.047	—	0.079	0.013	

Note: See Fig. 1 for population locations. Nonsignificant intraspecific pairwise differences (after Bonferroni correction) are presented in bold type. Values are significant at $p < 0.0033$ (river redhorse) and $p < 0.0019$ (shorthead redhorse).

Table 5. Summary of pairwise tests of genetic differentiation among Muskegon River river redhorse (*Moxostoma carinatum*; below diagonal) and shorthead redhorse (*Moxostoma macrolepidotum*; above diagonal) populations.

	MA	MC	MD	ME	MF	MG	MH
MA		**0.002**	0.014	0.009	0.008	0.015	**0.003**
MC	—		0.016	0.010	0.006	0.014	0.006
MD	—	—		0.007	0.013	0.010	0.008
ME	—	0.014	—		**0.005**	**0.002**	**0.003**
MF	—	0.010	—	**0.002**		0.012	0.004
MG	—	0.014	—	**<0.001**	**0.002**		**0.005**
MH	—	0.016	—	**0.005**	0.008	0.007	

Note: See Fig. 1 for population locations. Nonsignificant intraspecific pairwise differences (after Bonferroni correction) are presented in bold type. Values are significant at $p < 0.005$ (river redhorse) and $p < 0.0024$ (shorthead redhorse).

did not indicate that relatedness among individuals sampled was greater than that expected from a random population sample.

Population structure

Pairwise F_{ST} tests identified weak but statistically significant population structure in both rivers. Greater population subdivision was detected along the fragmented Trent River than in the Muskegon River (Tables 4 and 5): 86% of Trent River pairwise comparisons were significantly different, whereas fewer Muskegon River pairwise comparisons were significantly different (river redhorse, 60%; shorthead redhorse, 71%). Global F_{ST} values for Trent River redhorse populations were 1.5 times greater (shorthead redhorse) and five times greater (river redhorse) than for Muskegon River redhorse (Table 3). Only the difference between river redhorse global F_{ST} estimates was significantly different between rivers (randomization test: $p = 0.048$). For both Trent River species, neighbour-joining dendrograms (not shown) supported the level of differentiation identified by pairwise F_{ST} comparisons and indicated that population structuring reflects the geographic proximity of populations. Dendrograms of Muskegon River populations (not shown) indicated less structuring for river redhorse than in the Trent River but a similar pattern for shorthead redhorse (Reid 2007).

Along the Trent River, population differentiation was significantly greater for river redhorse than for shorthead redhorse (one-tailed paired t test: F_{ST}, $t = -1.93$, df = 14, $p = 0.04$; D_{CE}, $t = -4.02$, df = 14, $p < 0.001$). Alternatively, for the Muskegon River, there was no difference between the two redhorse species (one-tailed paired t test: F_{ST}, $t = -0.61$, df = 9, $p = 0.28$; D_{CE}, $t = -1.14$, df = 9, $p = 0.14$).

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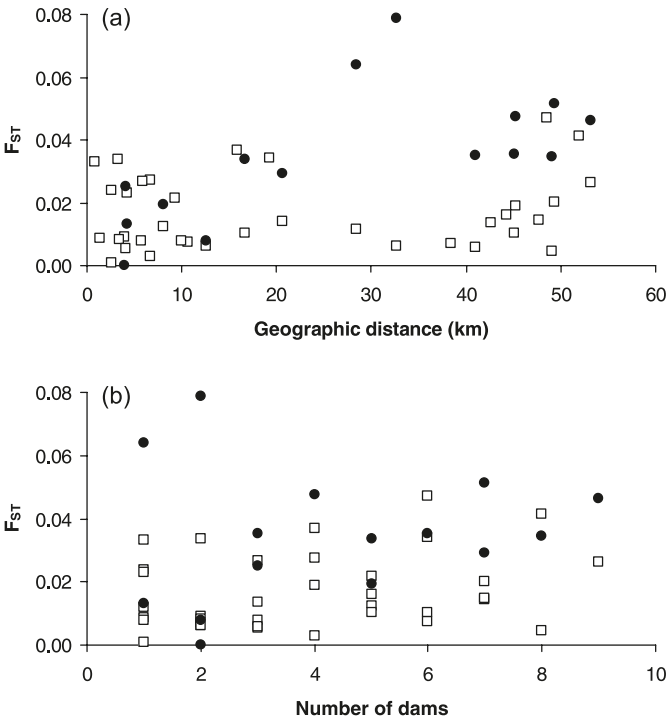
Fig. 2. Scatterplots of pairwise F_{ST} values versus (*a*) geographic distance and (*b*) number of dams separating pairs of Trent River populations of river redhorse (*Moxostoma carinatum*; solid circles) and shorthead redhorse (*Moxostoma macrolepidotum*; open squares).



An isolation-by-distance (IBD) relationship was identified for river redhorse population pairs along the Trent River (Mantel test: F_{ST}, $Z = 17.8$, $r = 0.63$, $p = 0.049$; D_{CE}, $Z = 139.9$, $r = 0.48$, $p = 0.05$), but not for the other species–river combinations. There were no significant relationships between pairwise F_{ST} and D_{CE} values and the number of intervening dams. The significant IBD relationship suggests a balance between gene flow and drift for river redhorse populations in the Trent River (Hutchison and Templeton 1999, case I). Alternatively, scatterplots of F_{ST} values against geographic distance (Figs. 2 and 3) suggest that Muskegon River redhorse populations and Trent River shorthead redhorse populations are not in regional equilibrium and that gene flow is more influential than drift (Hutchison and Templeton 1999, case I). In all iterations of the two runs (70 000 per run), *I* (parameter estimated by 2mod) indicated that a migration–drift equilibrium model was more likely than a non-equilibrium drift-only model.

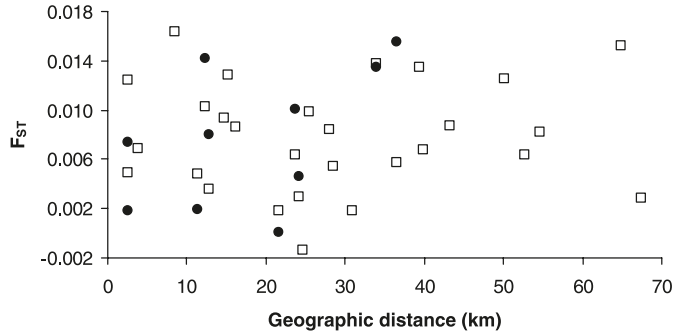
Individual assignment tests were unable to provide support for source–sink population dynamics along the Trent River. Despite population subdivision along the Trent River, only a small number of individual river redhorse or shorthead redhorse from hypothesized source populations were assigned back to their collection sites (Table 6). Accordingly, GeneClass was unable to assign individuals from putative sinks to hypothesized source populations.

Discussion

Genetic diversity and linkage disequilibrium

As predicted, redhorse genetic diversity in the fragmented

Fig. 3. Scatterplots of pairwise F_{ST} values versus geographic distance separating pairs of Muskegon River populations of river redhorse (*Moxostoma carinatum*; solid circles) and shorthead redhorse (*Moxostoma macrolepidotum*; open squares).



Trent River was lower than that along the free-flowing reach of the Muskegon River. However, there was little evidence of recent population bottlenecks, among-site differences in diversity, or a relationship between genetic diversity and river fragment length. River fragmentation effects on genetic diversity have not been consistent across studies. Relative to below-dam populations, lower genetic diversity has been measured in headwater populations of river sculpin (*Cottus gobio*) (Hanfling and Weetman 2006) and white-spotted charr (*Salvelinus leucomaenis*) isolated by dams (Yamamoto et al. 2004). Conversely, no differences were detected among fragmented populations of black redhorse (*Moxostoma duquesnei*) (Reid et al. 2008) and several salmonid species (Neraas and Spruell 2001; Heggenes and Roed 2006; Deiner et al. 2007). Differences among study findings likely reflect variation in the duration of isolation, river fragment length, permeability of dams to gene flow, and species susceptibility to fragmentation effects. Additionally, these studies did not compare fragmented and free-flowing rivers. Without Muskegon River data, interpretation of fragmentation effects on the genetic characteristics of Trent River populations would have been limited.

Along the Trent River, redhorse population size was positively correlated with fragment length. However, in contrast to theoretical expectations, genetic diversity was not lower in small fragments. Only indirect evidence of a fragment size effect was provided through the correlation of shorthead redhorse observed heterozygosity with spawning run size. A habitat fragment size effect on genetic diversity has been shown for Lahontan cutthroat trout (*Oncorhynchus clarkia henshawi*; Neville et al. 2006) but not other stream-dwelling salmonids (Yamamoto et al. 2004; Whiteley et al. 2006). Explanations for the absence of any relationship include the following: (*i*) fragmentation was not sufficiently severe or long enough, (*ii*) gene flow was high enough to limit loss of genetic variation, (*iii*) habitat size alone is not a good predictor of population size and therefore genetic diversity, and (*iv*) non-equilibrium population dynamics (i.e., recent population declines or expansions) (Yamamoto et al. 2004; Dhuyvetter et al. 2005; Whiteley et al. 2006). If fragmentation of the Trent River has produced source–sink population dynamics, the genetic character of sinks will mimic source populations when persistence times for sink populations are short and (or) immigration events are common (Gaggiotti and Smouse 1996).

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Table 6. Summary of the number of individual redhorse (river redhorse (*Moxostoma carinatum*); shorthead redhorse (*Moxostoma macrolepidotum*)) correctly re-assigned by GeneClass (Cornuet et al. 1999) to their putative Trent River source populations.

	Population						
Species	TA	TC	TF	TG	TH	TI	TJ
M. carinatum	2/41 (4.9%)	0/47 (0%)	0/47 (0%)	—	2/48 (4.2%)	0/56 (0%)	0/35 (0%)
M. macrolepidotum	2/48 (4.2%)	2/51 (3.9%)	0/54 (0%)	2/57 (3.5%)	1/53 (1.9%)	1/55 (1.8%)	0/30 (0%)

Note: See Fig. 1 for population locations. Numbers of individual are presented as fractions of local sample sizes (percentages are provided within parentheses).

Measures of genetic diversity for both redhorse species were similar to those reported for Sacramento sucker (*Catostomus occidentalis*) (Whitehead et al. 2003), copper redhorse (*M. hubbsi*) (Lippe et al. 2006), and black redhorse (*M. duquesnei*) (Reid et al. 2008) and greater than that reported by DeWoody and Avise (2000) for most freshwater fishes. The relative insensitivity of catostomids to factors hypothesized to negatively affect genetic diversity may be related to their relatively late age of maturity and long life span (Whitehead et al. 2003; Lippe et al. 2006). In the Trent River, river redhorse begin to mature at age 6 and live up to at least 27 years. Shorthead redhorse begin to mature at age 4 and live up to at least 19 years (Reid 2006, 2007). The effects of genetic drift or population decline are expected to increase with each generation. Therefore, over the same period, river redhorse and shorthead redhorse would be less affected by rapid population declines and isolation than fishes with shorter generation times and life spans (e.g., river sculpin; Hanfling and Weetman 2006).

Linkage disequilibrium at the total population level (all populations combined) in the fragmented Trent River was substantially greater than in the Muskegon River. Linkage disequilibrium has been predicted to occur as a result of inbreeding, population bottlenecks, population admixture, population subdivision, and extinction–recolonization population dynamics (Ohta 1982*a*, 1982*b*; McVean 2002). To date, only a few studies investigating habitat fragmentation or regional population structure have reported increased linkage disequilibrium (Stewart et al. 1999; Tero et al. 2003; Zartman et al. 2006). As population persistence in small Trent River fragments is dependent on emigration from larger fragments (i.e., source–sink dynamics), greater linkage disequilibrium could result from extinction–recolonization dynamics.

Population structure

Higher global F_{ST} values indicate greater population subdivision in the fragmented Trent River than along the free-flowing reach of the Muskegon River. However, pairwise genetic distance estimates were not correlated with the number of intervening dams, and F_{ST} values for Trent River population pairs were much lower than those reported for other fragmented riverine fish populations (Meldgaard et al. 2003; Yamamoto et al. 2004; Hanfling and Weetman 2006). The moderate effect of fragmentation is likely the result of navigational locks that facilitate some upstream and downstream movement and gene flow by redhorse (Reid 2007).

Evidence supporting a non-equilibrium population model (source–sink or classic metapopulation) to describe redhorse population dynamics along the Trent River was provided by linkage disequilibrium at the total population level (indicative of extinction–recolonization dynamics) and the absence of an isolation-by-distance relationship for shorthead redhorse (no migration–drift equilibrium). Non-equilibrium conditions have also been reported for sets of plant and insect populations linearly distributed along rivers (Tero et al. 2003; Dhuyvetter et al. 2005; Liu et al. 2006). Of the two non-equilibrium models, demographic data supports source–sink dynamics as population persistence in small river fragments is dependent on migrants from larger fragments. However, predicted genetic diversity differences between small and large fragments were not observed. Differences among Trent River fragments can be expected to be limited if persistence times for sink populations are short and (or) immigration events are common (Gaggiotti and Smouse 1996) and if several potential source populations all contribute to the genetic diversity of sinks (migrant pool mode; Slatkin 1977).

Individual assignment tests were not able to assign individuals sampled from hypothesized Trent River sink populations to potential source populations, or conclusively reassign individuals sampled from potential source populations back to their collection sites. The ability of Bayesian assignment methods to correctly identify migrants and the number of populations is affected by the level of gene flow among populations, the number of individuals sampled within each population, and the number of genotyped loci (Cornuet et al. 1999; Waples and Gaggiotti 2006). Global F_{ST} estimates for river redhorse (θ = 0.034) and shorthead redhorse (θ = 0.014) in the Trent River indicate low population differentiation. High assignment probabilities require either greater levels of population differentiation ($F_{ST} > 0.1$; Manel et al. 2002) or more loci than were genotyped in this study (13 loci (Berry et al. 2004); 20+ loci (Cornuet et al. 1999)).

Gene flow between source and sink populations is expected to be unidirectional. Instead, mark–recapture data indicate movement among small fragments and from small fragments to larger ones (Reid 2007). This pattern could explain the lack of difference in genetic diversity among fragments and the inability of assignment tests to assign individuals to hypothesized source populations. Alternatively, rather than being sink populations, small fragments may function as migratory corridors or matrix habitat (Fahrig 2003) between populations in larger fragments.

Species differences in fragmentation effects

As predicted, shorthead redhorse populations were characterized by greater genetic diversity and less differentiation than river redhorse. This pattern could result from species-level differences in population size, habitat specialization, and proximity to the core of the species range. During the spawning period, the number of shorthead redhorse (aver-

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aged across populations) captured during boat-electrofishing was two (Trent) and five (Muskegon) times greater than river redhorse (Reid 2007). For neutral DNA markers like microsatellites, the expected level of polymorphism is proportional to effective population size. Populations of abundant species are expected to have greater genetic diversity (Bazin et al. 2006), and the loss of alleles through genetic drift is most likely in small isolated populations (Frankham 1996; Yamamoto et al. 2004). Therefore, fragmented populations of the more abundant shorthead redhorse would be less susceptible than river redhorse to genetic drift and the corresponding increase in population differentiation.

Based on Wright's island model, it has been proposed that only a few migrants (~1–10 per generation) between subdivided populations are required to maintain genetic diversity (Mills and Allendorf 1996; Frankham et al. 2002). If the likelihood of movement by individual river redhorse and shorthead redhorse between fragments can be considered equal, a larger number of migrants would be exported from shorthead redhorse populations, increasing the chance of meeting the threshold number of migrants. As Wright's island model also predicts that larger populations require fewer migrants to arrest divergence than smaller ones (Wang 2004), the likelihood of divergence among shorthead redhorse populations would be further reduced.

As a result of reduced availability of preferred habitats or hosts, populations of specialist species are often smaller and more isolated than habitat generalists and, correspondingly, have lower neutral genetic variation (Zayed et al. 2005). River redhorse is primarily found in the large rivers of the Laurentian Great Lakes basin. In the Lake Ontario and Lake Michigan basins, it is only found in the Trent River and in the Fox, Grand, Muskegon, and St. Joseph rivers, respectively. Compared with shorthead redhorse, which is widely distributed across rivers and lakes in the Great Lakes basin (Mandrak and Crossman 1992; Hubbs and Lagler 2004), there is less opportunity to maintain genetic diversity through gene flow. Heithaus and Laushman (1997) found differences in genetic diversity among co-occurring riverine fishes to be related to ecological specialization and corresponding sensitivity to pollution events. Based on different associations with free-flowing riverine habitats, the magnitude of fragmentation effects was predicted to differ between redhorse species. However, the effect on genetic diversity was similar, and between-species differences in genetic diversity were measured in both rivers. Additionally, fragmentation had similar effects on movement among fragments and population demographics (Reid 2007). Therefore, species-level differences in genetic diversity and population subdivision are more likely the result of habitat specialization effects on population size than different demographic responses to river fragmentation.

Based on geographic isolation, greater temporal variability in population size, and smaller effective population sizes, species at the periphery of their geographic range are expected to have low within-population genetic diversity (Lesica and Allendorf 1995). Reduced diversity has been reported for Arctic grayling (*Thymallus arcticus arcticus*; Stamford and Taylor 2004) and bull trout (*Salvelinus confluentus*; Whiteley et al. 2006) populations at the extremes of their North American ranges. In this study, river redhorse populations are isolated and at the northern periphery of the species' range. Alternatively, shorthead redhorse is widely distributed across the Laurentian Great Lakes basin and close to the core of its geographic range (Mandrak and Crossman 1992; Scott and Crossman 1998). In regards to population size and proximity to other populations at a regional level, differences between core and peripheral populations and between habitat generalists and specialists are similar (Vucetich and Waite 2003; Zayed et al. 2005). In this study, the relative influence of either phenomenon on genetic diversity and population structure could not be separated as river redhorse is a habitat specialist at the northern edge of its range and shorthead redhorse is a habitat generalist near the core of its range.

Future research

In this study, redhorse life-history characteristics and the permeability of the dam and lock system likely mitigated fragmentation effects. A broader understanding of the genetic consequences of river fragmentation requires further research on (*i*) smaller-bodied fishes with more limited dispersal patterns and shorter generation times (e.g., longnose dace, *Rhinichthys cataractae*), (*ii*) rivers fragmented by large hydroelectric or flood-control dams that present more substantial barriers to movement, and (*iii*) the effect of restored connectivity (e.g., dam removal) on previously fragmented populations. The inclusion of demographic and movement data and reference populations in future studies will support more conclusive inference (McElroy et al. 2003; Wilson et al. 2004; Johansson et al. 2006) and reduce the likelihood of false interpretations of fragmentation effect (Whitehead et al. 2003).

Acknowledgments

This study was supported by Fisheries and Oceans Canada, Endangered Species Recovery Fund (WWF Canada and Environment Canada), Federal Departmental Recovery Fund, Ontario Ministry of Natural Resources, Ontario Power Generation, Parks Canada, and the US Great Lakes Fish and Wildlife Restoration Act. S. Reid was supported by Natural Sciences and Engineering Research Council of Canada (NSERC) Industrial and Ontario Graduate scholarships. P. Addison, J. Barnucz, A. Edwards, H. Gignac, N. Koutrilides, J. MacLeod, and C. Sinclair assisted with field sampling. D. Gillette and M. Lavender provided valuable assistance during laboratory data collection. The manuscript was greatly improved by comments provided by N. Mandrak, J. Schaefer, W. Stott, and three anonymous reviewers.

References

Allendorf, F.W. 1986. Genetic drift and the loss of alleles versus heterozygosity. Zoo Biol. **5**: 181–190. doi:10.1002/zoo.1430050212.

Angus, J.T. 1998. A respectable ditch: a history of the Trent–Severn Waterway, 1833–1920. McGill–Queen's University Press, Kingston, Ont.

Bazin, E., Glemin, S., and Galtier, N. 2006. Population size does not influence mitochondrial genetic diversity in animals. Science (Washington, D.C.), **312**: 570–572. doi:10.1126/science.1122033. PMID:16645093.

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Belkhir, K., Castric, V., and Bonhomme, F. 2002. IDENTIX, a software to test for relatedness in a population using permutation methods. Mol. Ecol. Notes, **2**: 611–614. doi:10.1046/j.1471-8286.2002.00273.x.

Bermingham, E., and Moritz, C. 1998. Comparative phylogeography: concepts and applications. Mol. Ecol. **7**: 367–369. doi:10.1046/j.1365-294x.1998.00424.x.

Berry, O., Tocher, M.D., and Sarre, S.D. 2004. Can assignment tests measure dispersal? Mol. Ecol. **13**: 551–561. doi:10.1046/j.1365-294X.2004.2081.x. PMID:14871360.

Bohonak, A.J. 2002. IBD (isolation by distance): a program for analyses of isolation by distance. J. Hered. **93**: 153–154. doi:10.1093/jhered/93.2.153. PMID:12140277.

Castric, V., and Bernatchez, L. 2004. Individual assignment test reveals differential restriction to dispersal between two salmonids despite no increase of genetic differences with distance. Mol. Ecol. **13**: 1299–1312. doi:10.1111/j.1365-294X.2004.02129.x. PMID:15078465.

Cavalli-Sforza, L.L., and Edwards, W.W.F. 1967. Phylogenetic analysis: models and estimation procedures. Evolution, **21**: 550–570. doi:10.2307/2406616.

Chen, Y., and Harvey, H. 1999. Spatial structuring of length-at-age of the benthivorous white sucker (*Catostomus commersoni*) in relation to environmental variables. Aquat. Living Resour. **12**: 351–362. doi:10.1016/S0990-7440(99)00108-4.

Ciofi, C., Beaumont, M.A., Swingland, I.R., and Bruford, M.W. 1999. Genetic divergence and units for conservation in the Komodo dragon *Varanus komodoensis*. Proc. R. Soc. Lond. Ser. B Biol. Sci. **266**: 2269–2274. doi:10.1098/rspb.1999.0918.

Committee on the Status of Endangered Wildlife in Canada. 2005. Canadian species at risk. Committee on the Status of Endangered Wildlife in Canada, Ottawa, Ont.

Cornuet, J.M., and Luikart, G. 1996. Description and power analysis of two tests for detecting recent population bottlenecks from allele frequency data. Genetics, **144**: 2001–2014. PMID:8978083.

Cornuet, J.-M., Piry, S., Luikart, G., Estoup, A., and Solignac, M. 1999. New methods employing multi-locus genotypes to select or exclude populations as origins of individuals. Genetics, **153**: 1987–2000.

Cushman, C.A. 2006. Effects of habitat loss and fragmentation on amphibians: a review and prospectus. Biol. Conserv. **128**: 231–240. doi:10.1016/j.biocon.2005.09.031.

Debinski, D.M., and Holt, R.D. 2000. Review: a survey and overview of habitat fragmentation experiments. Conserv. Biol. **14**: 342–355. doi:10.1046/j.1523-1739.2000.98081.x.

Deiner, K., Garza, J.C., Coey, R., and Girman, D.J. 2007. Population structure and genetic diversity of trout (*Onchorynchus mykiss*) above and below natural and man-made barriers in the Russian River, California. Conserv. Genet. **8**: 437–454. doi:10.1007/s10592-006-9183-0.

DeWoody, J.A., and Avise, J.C. 2000. Microsatellite variation in marine, freshwater and anadromous fishes compared to other animals. J. Fish Biol. **56**: 461–473. doi:10.1111/j.1095-8649.2000.tb00748.x.

Dhuyvetter, H., Gaublomme, E., and Desender, K. 2005. Bottlenecks, drift and differentiation: the fragmented population structure of the saltmarsh beetle *Pogonus chalceas*. Genetica, **124**: 167–177. doi:10.1007/s10709-005-1157-5. PMID:16134330.

Ewers, R.M., and Didham, R.K. 2006. Confounding factors in the detection of species responses to habitat fragmentation. Biol. Rev. Camb. Philos. Soc. **81**: 117–142. PMID:16318651.

Fahrig, L. 2003. Effects of habitat fragmentation on biodiversity. Annu. Rev. Ecol. Evol. Syst. **34**: 487–515. doi:10.1146/annurev.ecolsys.34.011802.132419.

Frankham, R. 1996. Relationship of genetic variation to population size in wildlife. Conserv. Biol. **10**: 1500–1508. doi:10.1046/j.1523-1739.1996.10061500.x.

Frankham, R., Ballou, J.D., and Briscoe, D.A. 2002. Introduction to conservation genetics. Cambridge University Press, Cambridge, UK.

Gaggiotti, O.E., and Smouse, P.E. 1996. Stochastic migration and maintenance of genetic variation in sink populations. Am. Nat. **147**: 919–945. doi:10.1086/285886.

Gaggiotti, O.E., Lange, O., Rassmann, K., and Glidden, C. 1999. A comparison of two indirect methods for estimating gene flow using microsatellite data. Mol. Ecol. **8**: 1513–1520. doi:10.1046/j.1365-294x.1999.00730.x. PMID:10564457.

Garza, J.C., and Williamson, E.G. 2001. Detection of reduction in population size using data from microsatellite loci. Mol. Ecol. **10**: 305–318. doi:10.1046/j.1365-294x.2001.01190.x. PMID:11298947.

Goudet, J. 2001. FSTAT, a program to estimate and test gene diversities and fixation indices (version 2.9.3). Available from http://www2.unil.ch/popgen/softwares/fstat.htm.

Grabowski, T.B., and Isely, J.J. 2006. Seasonal and diel movements and habitat use of robust redhorse in the lower Savannah River, Georgia and South Carolina. Trans. Am. Fish. Soc. **135**: 1145–1155. doi:10.1577/T05-230.1.

Hackney, P.A., Tatum, W.M., and Spencer, S.L. 1968. Life history study of the river redhorse *Moxostoma carinatum* in the Cahaba River, Alabama, with notes on the management of the species as a sport fish. Proc. Annu. Conf. Southeast. Assoc. Game Fish Comm. **21**: 324–332.

Hanfling, B., and Weetman, D. 2006. Concordant genetic estimators of migration reveal anthropogenically enhanced source–sink population structure in the river sculpin, *Cottus gobio*. Genetics, **173**: 1487–1501. doi:10.1534/genetics.105.054296. PMID:16624916.

Harrison, S. 1991. Local extinction in a metapopulation context: an empirical evaluation. Biol. J. Linn. Soc. **42**: 73–88. doi:10.1111/j.1095-8312.1991.tb00552.x.

Heggenes, J., and Roed, K.H. 2006. Do dams increase genetic diversity in brown trout (*Salmo trutta*)? Microgeographic differentiation in a fragmented river. Ecol. Freshw. Fish, **15**: 366–375. doi:10.1111/j.1600-0633.2006.00146.x.

Heithaus, M.R., and Laushman, R.H. 1997. Genetic variation and conservation of stream fishes: influence of ecology, life history and water quality. Can. J. Fish. Aquat. Sci. **54**: 1822–1836. doi:10.1139/cjfas-54-8-1822.

Hernandez-Martich, J.D., and Smith, M.H. 1997. Downstream gene flow and genetic structure of *Gambusia holbrooki* (eastern mosquitofish) populations. Heredity, **79**: 295–301.

Hubbs, C.L., and Lagler, K.F. 2004. Fishes of the Great Lakes region. *Revised by* G.R. Smith. University of Michigan Press, Ann Arbor, Mich.

Hutchison, D.W., and Templeton, A.R. 1999. Correlation of pairwise genetic and geographic distance measures: inferring the relative influences of gene flow and drift on the distribution of genetic variability. Evolution, **53**: 1898–1914. doi:10.2307/2640449.

Jenkins, R.E., and Burkhead, N.M. 1993. Freshwater fishes of Virginia. American Fisheries Society, Bethesda, Md.

Johansson, M., Primmer, C.R., and Merilä, J. 2006. History vs. current demography: explaining the genetic population structure of the common frog (*Rana temporaria*). Mol. Ecol. **15**: 975–983. PMID:16599961.

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Lesica, P., and Allendorf, F.W. 1995. When are peripheral populations valuable for conservation? Conserv. Biol. **9**: 753–760. doi:10.1046/j.1523-1739.1995.09040753.x.

Lippe, C., Dumont, P., and Bernatchez, L. 2004. Isolation and identification of 21 microsatellite loci in copper redhorse (*Moxostoma hubbsi*; Catostomidae) and their variability in other catostomids. Mol. Ecol. Notes, **4**: 638–641. doi:10.1111/j.1471-8286.2004.00758.x.

Lippe, C., Dumont, P., and Bernatchez, L. 2006. High genetic variability and no inbreeding in the endangered copper redhorse, *Moxostoma hubbsi* (Catostomidae, Pisces): the positive sides of a long generation time. Mol. Ecol. **15**: 1769–1780. doi:10.1111/j.1365-294X.2006.02902.x. PMID:16689897.

Liu, Y., Wang, Y., and Huang, H. 2006. High interpopulation genetic differentiation and unidirectional linear migration patterns in *Myricaria laxiflora* (Tamaricaceae), an endemic riparian plant in the Three Gorges Valley of the Yantze River. Am. J. Bot. **93**: 206–215. doi:10.3732/ajb.93.2.206.

Luikart, G., and Cornuet, J.-M. 1998. Empirical evaluation of a test for identifying recently bottlenecked populations from allele frequency data. Conserv. Biol. **12**: 228–237. doi:10.1046/j.1523-1739.1998.96388.x.

Mandrak, N.E., and Crossman, E.J. 1992. A checklist of Ontario freshwater fishes. Royal Ontario Museum, Toronto, Ont.

Manel, S., Berthier, P., and Luikart, G. 2002. Detecting wildlife poaching: identifying the origin of individuals with Bayesian assignment tests and multi-locus genotypes. Conserv. Biol. **16**: 650–659. doi:10.1046/j.1523-1739.2002.00576.x.

McElroy, T.C., Kandl, K.L., Garcia, J., and Trexler, J.C. 2003. Extinction–colonization dynamics structure genetic variation of spotted sunfish (*Lepomis punctatus*) in the Florida Everglades. Mol. Ecol. **12**: 355–368. doi:10.1046/j.1365-294X.2003.01738.x. PMID:12535087.

McVean, G.A.T. 2002. A genealogical interpretation of linkage disequilibrium. Genetics, **162**: 987–991. PMID:12399406.

Meldgaard, T., Nielson, E.E., and Loeschcke, V. 2003. Fragmentation by weirs in a riverine system: a study of genetic variation in time and space among populations of European grayling (*Thymallus thymallus*) in a Danish river system. Conserv. Genet. **4**: 735–747. doi:10.1023/B:COGE.0000006115.14106.de.

Mills, L.S., and Allendorf, F.W. 1996. The one-migrant per generation rule in conservation and management. Conserv. Biol. **10**: 1509–1518. doi:10.1046/j.1523-1739.1996.10061509.x.

NatureServe. 2005. NatureServe Explorer: an online encyclopedia of life. Version 3.0. Available from www.natureserve.org [accessed 15 November 2005].

Neraas, L.P., and Spruell, P. 2001. Fragmentation of riverine systems: the genetic effects of dams on bull trout (*Salvelinus confluentus*) in the Clark Fork River system. Mol. Ecol. **10**: 1153–1164. doi:10.1046/j.1365-294X.2001.01269.x. PMID:11380874.

Neville, H.M., Dunham, J.B., and Peacock, M.M. 2006. Landscape attributes and life history variability shape genetic structure of trout populations in a stream network. Landsc. Ecol. **21**: 901–916. doi:10.1007/s10980-005-5221-4.

Ohta, T. 1982*a*. Linkage disequilibrium with the island model. Genetics, **101**: 139–155. PMID:17246079.

Ohta, T. 1982*b*. Linkage disequilibrium due to random genetic drift in finite subdivided populations. Proc. Natl. Acad. Sci. U.S.A. **79**: 1940–1944. doi:10.1073/pnas.79.6.1940. PMID:16593171.

Paetkau, D., Slade, R., Burden, M., and Estoup, A. 2004. Genetic assignment methods for the direct, real-time estimation of migration rate: a simulation-based exploration of accuracy and power. Mol. Ecol. **13**: 55–65. doi:10.1046/j.1365-294X.2004.02008.x. PMID:14653788.

Page, R.D.M. 1996. TREEVIEW: an application to display phylogenetic trees on personal computers. Comput. Appl. Biosci. **12**: 357–358. PMID:8902363.

Park, S.D.E. 2001. Trypanotolerance in West African cattle and the population genetic effects of selection. Ph.D. thesis, Department of Genetics, University of Dublin, Dublin, Eire.

Peakall, R., and Smouse, P.E. 2006. GenAlEx 6: genetic analysis in Excel. Population genetic software for teaching and research. Mol. Ecol. Notes, **6**: 288–295. doi:10.1111/j.1471-8286.2005.01155.x.

Piry, S., Alapetite, A., Cornuet, J.-M., Paetkau, D., Baudouin, L., and Estoup, A. 2004. GeneClass2: a software for genetic assignment and first-generation migrant detection. J. Hered. **95**: 536–539. doi:10.1093/jhered/esh074. PMID:15475402.

Piry, S., Luikart, G., and Cornuet, J. 1999. BOTTLENECK: a computer program for detecting recent reductions in effective population size using allele frequency data. J. Hered. **90**: 502–503. doi:10.1093/jhered/90.4.502.

Pringle, C.M., Freeman, M.C., and Freeman, B.J. 2000. Regional effects of hydrologic alterations on riverine macrobiota in the New World: tropical–temperate comparisons. Bioscience, **50**: 807–823. doi:10.1641/0006-3568(2000)050[0807:REOHAO]2.0.CO;2.

Queller, D.C., and Goodnight, K.F. 1989. Estimating relatedness using genetic markers. Evolution, **43**: 258–275. doi:10.2307/2409206.

Quinn, J.W., and Kwak, T.J. 2003. Fish assemblage changes in an Ozark river after impoundment: a long-term perspective. Trans. Am. Fish. Soc. **132**: 110–119. doi:10.1577/1548-8659(2003)132<0110:FACIAO>2.0.CO;2.

Rannala, B., and Mountain, J.L. 1997. Detecting immigration by using multilocus genotypes. Proc. Natl. Acad. Sci. U.S.A. **94**: 9197–9201. doi:10.1073/pnas.94.17.9197. PMID:9256459.

Raymond, M., and Rousset, F. 1995. GENEPOP (version 1.2): population genetics software for exact test and ecumenicism. J. Hered. **86**: 248–249.

Reid, S.M. 2006. Timing and demographic characteristics of redhorse spawning runs in three Great Lakes basin rivers. J. Freshw. Ecol. **21**: 249–258.

Reid, S.M. 2007. Effects of river fragmentation on the distribution, demographics and genetic characteristics of redhorse (*Moxostoma* spp.) populations. Ph.D. thesis, Watershed Ecosystems Graduate Program, Trent University, Peterborough, Ontario.

Reid, S.M., Wilson, C.C., Mandrak, N.E., and Carl, L.M. 2007. Population structure and genetic diversity of black redhorse (*Moxostoma duquesnei*) in a highly fragmented watershed. Conserv. Genet.. **9**: 531–546. doi:10.1007/s10592-007-9367-2.

Rice, W.R. 1989. Analyzing tables of statistical tests. Evolution, **43**: 223–225. doi:10.2307/2409177.

Rousset, F. 1997. Genetic differentiation and estimation of gene flow from *F*-statistics under isolation by distance. Genetics, **145**: 1219–1228. PMID:9093870.

Scott, W.B., and Crossman, E.J. 1998. Freshwater fishes of Canada. Galt House Publications Ltd., Oakville, Ont.

Slatkin, M. 1977. Gene flow and genetic drift in a species subject to frequent local extinctions. Theor. Popul. Biol. **12**: 253–262. doi:10.1016/0040-5809(77)90045-4. PMID:601717.

Srikwan, S., and Woodruff, D.S. 2000. Genetic erosion in isolated small mammal populations following rain forest fragmentation. *In* Genetics, demography and viability of fragmented populations. *Edited by* A. Young and G. Clarke. Cambridge University Press, Cambridge, UK. pp. 149–172.

Stamford, M.D., and Taylor, E.B. 2004. Phylogeographical lineages of Arctic grayling (*Thymallus arcticus*) in North America: diver-

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gence, origins, and affinities with Eurasian *Thymallus*. Mol. Ecol. **13**: 1533–1549. doi:10.1111/j.1365-294X.2004.02174.x. PMID:15140096.

Stewart, W.A., Dallas, J.F., Piertney, S.B., Marshall, F., Lambin, X., and Telfer, S. 1999. Metapopulation genetic structure in the water vole, *Arvicola terrstris*, in northeastern Scotland. Biol. J. Linn. Soc. **68**: 159–171.

Takezaki, N., and Nei, M. 1996. Genetic distance and reconstruction of phylogenetic trees from microsatellite DNA. Genetics, **144**: 389–399. PMID:8878702.

Tero, N., Aspi, J., Siikamaki, P., Jakalaniem, A., and Tuomi, J. 2003. Genetic structure and gene flow in a metapopulation of an endangered species, *Silene tatarica*. Mol. Ecol. **12**: 2073–2085. doi:10.1046/j.1365-294X.2003.01898.x. PMID:12859630.

Tranah, G.J., Agresti, J.J., and May, B. 2001. New microsatellite loci for suckers (Catostomidae): primer homology in *Catostomus*, *Chasmistes* and *Deltistes*. Mol. Ecol. Notes, **1**: 55–60. doi:10.1046/j.1471-8278.2000.00023.x.

Trippel, E.A., and Harvey, H.H. 1987. Abundance, growth and food supply of white sucker (*Catostomus commersonii*) in relation to lake morphometry and pH. Can. J. Zool. **65**: 558–567. doi:10.1139/z87-086.

Vallone, P.M., and Butler, J.M. 2004. AutoDimer: a screening tool for primer–dimer and hairpin structures. Biotechniques, **37**: 226–231. PMID:15335214.

van Oosterhout, C., Hutchinson, W.F., Willis, D.P.M., and Shipley, P. 2004. MICROCHECKER: software for identifying and correcting genotyping errors in microsatellite data. Mol. Ecol. Notes, **4**: 535–538. doi:10.1111/j.1471-8286.2004.00684.x.

Vucetich, J.A., and Waite, T.A. 2003. Spatial patterns of demography and genetic processes across the species' range: null hypotheses for landscape conservation genetics. Conserv. Genet. **4**: 639–645. doi:10.1023/A:1025671831349.

Wang, J. 2004. On the application of the one-migrant-per-generation rule to conservation and management. Conserv. Biol. **18**: 332–343. doi:10.1111/j.1523-1739.2004.00440.x.

Waples, R.S. 2002. Definition and estimation of effective population size in the conservation of endangered species: *In* Population viability analysis. *Edited by* S.R. Beissinger and D.R. McCullough. University of Chicago Press, Chicago, Ill. pp. 147–168.

Waples, R.S., and Gaggiotti, O. 2006. What is a population? An empirical evaluation of some genetic methods for identifying the number of gene pools and their degree of connectivity. Mol. Ecol. **15**: 1419–1439. doi:10.1111/j.1365-294X.2006.02890.x. PMID:16629801.

Weir, B.S., and Cockerham, C.C. 1984. Estimating *F*-statistics for the analysis of population structure. Evolution, **38**: 1358–1370. doi:10.2307/2408641.

Whitehead, A., Anderson, S.L., Kuivila, K.M., Roach, J.L., and May, B. 2003. Genetic variation among interconnected populations of *Catostomus occidentalis*: implications for distinguishing impacts of contaminants from biogeographic structuring. Mol. Ecol. **12**: 2817–2833. doi:10.1046/j.1365-294X.2003.01933.x. PMID:12969484.

Whiteley, A.R., Spruell, P., Rieman, B.E., and Allendorf, F.W. 2006. Fine-scale genetic structure of bull trout at the southern limit of their distribution. Trans. Am. Fish. Soc. **135**: 1238–1253. doi:10.1577/T05-166.1.

Whiteley, A.R., Spruell, P., and Allendorf, F.W. 2004. Ecological and life history characteristics predict population genetic divergence of two salmonids in the same landscape. Mol. Ecol. **13**: 3675–3688. doi:10.1111/j.1365-294X.2004.02365.x. PMID:15548282.

Wilson, A.J., Hutchings, J.A., and Ferguson, M.M. 2004. Dispersal in a stream-dwelling salmonids: inference from tagging and microsatellite studies. Conserv. Genet. **5**: 25–37. doi:10.1023/B:COGE.0000014053.97782.79.

Yamamoto, S., Morita, K., Koizumi, I., and Maekawa, K. 2004. Genetic differentiation of white-spotted charr (*Salvelinus leucomaenis*) populations after habitat fragmentation: spatial–temporal changes in gene frequencies. Conserv. Genet. **5**: 529–538. doi:10.1023/B:COGE.0000041029.38961.a0.

Yoder, C.O., and Beaumier, R.A. 1986. The occurrence and distribution of river redhorse, *Moxostoma carinatum*, and greater redhorse, *Moxostoma valenciennesi*, in the Sandusky River, Ohio. Ohio J. Sci. **86**: 18–21.

Zartman, C.E., McDaniel, S.F., and Shaw, A.J. 2006. Experimental habitat fragmentation increases linkage disequilibrium but does not affect genetic diversity or population structure in the Amazonian liverwort *Radula flaccida*. Mol. Ecol. **15**: 2305–2315. doi:10.1111/j.1365-294X.2006.02929.x. PMID:16842407.

Zayed, A., Packer, L., Grixti, J.C., Owen, R.E., and Toro, H. 2005. Increased genetic differentiation in a specialist versus a generalist bee: implications for conservation. Conserv. Genet. **6**: 1017–1026. doi:10.1007/s10592-005-9094-5.

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